

Danh Tran

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EDUCATION

California State Polytechnic University, Pomona

Expected May 2029

Bachelor of Science in Mechanical Engineering, GPA: 3.99/4.00

SKILLS

Electrical: PCB Design, Altium, KiCAD, ESP32, Arduino, Raspberry Pi, CAN Bus, I2C, SPI, UART, BLE, RFID

Software: Python, C++, Bash, Java, JavaScript, HTML, CSS, React, ThreeJS, MATLAB, ROS, Git, Excel

Mechanical: SolidWorks, Autodesk Fusion, Generative Design, FEA, 3D Printing, Electromechanical Integration

EXPERIENCE

Robotics Engineer Research Intern | KiCAD, MATLAB, Autodesk Fusion, SolidWorks

May 2026 – August 2026

CREST-MECIS Robotics Research Lab

Edinburg, TX

- Co-designed a UAV with a modular 3D-printed frame, cutting total frame weight ~60% through generative design topology studies and stress testing in Autodesk Fusion
- Validated a 15 lb payload capacity and 20 to 30 minute unloaded flight endurance for the UAV platform
- Designed hardware schematics and PCB layouts in KiCAD for a UAV-deployed structural crack detection sensor
- Improved crack detection accuracy to 1 mm with ~0.5 mm uncertainty by analyzing laser crack sensor data from multiple 3D-printed samples in MATLAB

Robotics Engineer Research Intern | Python, C++, MATLAB, ROS, Motion Planning

December 2025 – May 2026

CREST Grant Robotics Research Lab

Pomona, CA

- Reduced distal mass and power draw by comparing degree of freedom (DoF) actuators, pneumatic systems, and compliant mechanisms and modeling gas forces in MATLAB
- Created a computer vision hand tracking system in Python that replicated human hand gestures on a robotic hand for teleoperation and human-robot interaction research
- Developed robotic hand control software in Python to record, store, and replay grasp configurations
- Built a cross-platform ROS bridge between an Artus Lite robotic hand on Linux and a HaptX haptic glove on Windows, enabling bidirectional teleoperation and force feedback
- Evaluated pneumatic grippers from multiple industrial vendors on gripping force, compliance, integration, and cost performance for EPSON manipulation applications
- Saved ~\$1,000 by sourcing missing robotic manufacturing parts, comparing in-house SolidWorks design and machining against market-ready parts, and negotiating vendor discounts
- Developed grasp selection by evaluating AI- and ML-based grasping across tactile sensors measuring force, pressure, and slip against cameras with point clouds, generalizing to unseen objects
- Integrated electromechanical hardware by studying CAN and UART datasheets and sizing components in SolidWorks, producing a system future researchers can modify
- Programmed multiple EPSON robots to run simultaneously in a manufacturing cell, writing lesson plans for an undergraduate robotic manufacturing class and documentation for a research paper
- Coordinated technical meetings between Sarcomere Dynamics and the lab, expediting replacement of a defective robotic hand to keep research on schedule

PROJECTS

Automated Powder Blender | SolidWorks, Mechanical Design, Chain Drive Systems

- Saved ~\$600 to \$700 versus commercial systems by designing a motorized drive for a ~\$400 manual powder blender that rotates a ~9 kg drum at ~30 RPM
- Validated motor selection through torque and rotational dynamics analysis of moment of inertia, angular acceleration, and efficiency losses, driving the system with a ~0.6 N·m motor on a 1:1 chain transmission

- Designed a chain drive of sprockets, tensioner, and housing in SolidWorks with 3D printed components, holding power transmission under a ~500 N estimated chain load

Acoustic Attenuation Device | SolidWorks, Thermodynamics, 3D Printing

- Designed a passive acoustic attenuation system for a pneumatic airsoft platform using wave interference
- Reduced acoustic signature ~10% by modeling compressible gas flow with adiabatic relations and ideal gas energy equations to redistribute peak pressure over time
- Optimized internal baffle geometry and chamber spacing through quarter-wavelength and Mach flow analysis, improving destructive interference and turbulence dissipation
- Cut post-processing time by ~5 minutes per unit by designing a support-free 3D-printable part with a 45-degree maximum overhang in ABS at 1.6 mm wall thickness

Pickleball Launcher | Raspberry Pi, ESP32, Python, C++, BLE, 3D Printing

- Scoped a pickleball training launcher with a practicing engineer, using Raspberry Pi AI cameras to detect players and adjust shot difficulty to skill level
- Cut ~\$200 in electronics and lowered energy use by comparing 3D-printed gears against capstan drives to optimize the ball feed rate and increase durability
- Developed the physical architecture in SolidWorks to fit the full mechanism in a 40x40x80 cm envelope, ~200% smaller than most commercial pickleball machines
- Built a dual motor shooter on Raspberry Pi in Python, then migrated it to ESP32 in C++ with H-bridge drivers, lowering energy use and adding BLE remote control through nRF Connect

RFID Jukebox | Raspberry Pi, Arduino, Python, C++, SolidWorks, 3D Printing

- Recreated a Minecraft jukebox as a working device that plays music from physical discs, covering mechanical design, embedded programming, and 3D printing
- Created a 50% thinner modular system and raised high-speed disc ejection success 150% by comparing push-push mechanisms and improving existing SolidWorks models for 3D printing
- Cut cost by \$40 by benchmarking Raspberry Pi in Python against Arduino in C++ with RFID components, mapping each RFID tag to an mp4 track across 2 speakers

Go-Kart | Altium, SolidWorks, Excel, Electromechanical Integration

- Built a working go-kart from a repurposed bedframe and leftover materials with a 4-member team
- Managed a \$1.2k budget and the schedule for 4 teammates and drivers, tracking part costs in Excel
- Designed a go-kart frame in SolidWorks supporting 200+ lbs of riders plus ~25 lbs of electronics, applying statics analysis and accounting for real fabrication tolerances
- Reduced distal mass and steering effort by applying torque and friction analysis to drivetrain troubleshooting and electrical wiring, delivering a functioning electrical system

Portfolio Website | HTML, CSS, JavaScript, React, ThreeJS, Cloudflare

- Built an interdisciplinary portfolio site showcasing mechanical, electrical, and software projects
- Designed a documentation page for each project in HTML and CSS, adding project visuals, 3D models, and interactive animations built with JavaScript, React, and ThreeJS

EXTRACURRICULARS

CPP Autonomous Vehicle Laboratory | C++, Altium, CAN Bus, I2C, SPI, UART, BLE 2026

November 2025 – January

Mechatronics Engineer Subteam Member

Pomona, CA

- Increased steering, throttle, and brake data collected for the software team by 130% by implementing master CAN node logic to request and store real-time actuator feedback
- Optimized Arduino and PCB communication by rewriting existing code in C++, replacing an unoptimized I2C, SPI, UART, and BLE stack with CAN bus infrastructure
- Upgraded a legacy autonomous vehicle design toward Level 5 autonomy with an AI camera system

- Led a self-taught 6-member team for 3 years from beginner to #1 in California Gold Tier, advancing to Platinum Tier and placing top 100 National Semifinalist twice, top 0.5% of 32,000 competitors nationwide
- Planned biweekly meetings to study and harden 8 government systems, including Windows desktop and Server, Linux distributions Ubuntu, Mint, Fedora, and Debian, and Cisco

CERTIFICATIONS AND HONORS

Certifications: State Seal of Civic Engagement, Seal of Biliteracy, Computer Science and Cyber Security CTE Pathway

Honors: Dean's List, AP Scholar with Distinction, National Honor Society, California Scholarship Federation